MATH 285J: Topics in High-Dimensional Sampling and Generative Diffusions

Lecture Notes (please use carefully; any typos let us know!)

Yifan Chen
UCLA, Department of Mathematics

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1 Course Goals

The central problem of this course is: *how do we sample from a target distribution $\pi \propto e^{-V}$ on $\mathbb{R}^d$ and compute statistics $\mathbb{E}_\pi[h]$, efficiently in high dimensions?* We assume access to π either via (1) **query access** to V or ∇V (with unknown normalization $Z = \int e^{-V}\,dx$), or (2) **data access** via existing samples $X_1, \ldots, X_N \sim \pi$. Motivating settings include:

- **Bayesian inverse problems**: given $y = F(\theta) + \xi$ where $\xi \sim \mathcal{N}(0, \sigma^2 I_d)$ and a prior $p_{\text{prior}}(\theta)$, sample the posterior $p_{\text{post}}(\theta) \propto p_{\text{prior}}(\theta)\, e^{-\|y-F(\theta)\|^2/(2\sigma^2)}$.
- **Statistical physics**: Gibbs measure $\pi \propto e^{-V/T}$ with potential V and temperature T; partition function $Z = \int e^{-V/T}\,dx$ is intractable.
- **Generative modeling**: π is an unknown data distribution accessible only through samples.

Course outline and references

- **Classical methods**: rejection/importance sampling, curse of dimensionality.
- **Dynamical methods**: MCMC, Langevin SDEs, mixing time and bias.
- **Flow and diffusion methods**: diffusion models, flow matching (ODEs/SDEs), gradient flows, variational inference, posterior sampling with generative priors.

References: Sanz-Alonso & Al-Ghattas (Intro to Monte Carlo); Chewi (Algorithmic Complexity of Log-concave Sampling); Liu (Monte Carlo Strategies in Scientific Computing); Sanz-Alonso, Stuart & Taeb (Inverse Problems and Data Assimilation); various diffusion/flow papers.

2 Rejection and Importance Sampling

2.1 Generating Samples in One Dimension

We assume throughout that $U \sim \mathrm{Unif}(0,1)$ can be generated. This suffices to sample any one-dimensional distribution via the following standard fact.

Proposition 2.1 (Inverse CDF method)**.** *Let π be a probability distribution on $\mathbb{R}$ with CDF $F(x) = \mathbb{P}_\pi(X \le x)$. If $U \sim \mathrm{Unif}(0,1)$, then $X = F^{-1}(U) \sim \pi$, where $F^{-1}(u) = \inf\{x : F(x) \ge u\}$.*

Proof. $\mathbb{P}(F^{-1}(U) \le x) = \mathbb{P}(U \le F(x)) = F(x)$. □

This gives Gaussian, exponential, and other standard distributions. For general high-dimensional targets, the CDF is intractable and more powerful methods are needed.

2.2 Rejection Sampling

Setup. Let $\pi, g : \mathbb{R}^d \to [0, \infty)$ be probability densities with $\mathrm{supp}(\pi) \subseteq \mathrm{supp}(g)$, and fix $M \geq \sup_{x \in \mathbb{R}^d} \pi(x)/g(x) < \infty$.

Algorithm. Repeat until acceptance:

(1) Draw $Y \sim g$ and $U \sim \mathrm{Unif}(0, 1)$ independently.

(2) Compute $r(Y) = \pi(Y)/(Mg(Y)) \in [0, 1]$.

(3) If $U \leq r(Y)$, return $X = Y$. Otherwise go to (1).

Proposition 2.2 (Correctness)**.** *The returned sample satisfies $X \sim \pi$.*

Proof. For any measurable $A \subseteq \mathbb{R}^d$,

$$\mathbb{P}(Y \in A \mid Y \text{ accepted}) = \frac{\int_A g(y)\, r(y)\, dy}{\int_{\mathbb{R}^d} g(y)\, r(y)\, dy} = \frac{\frac{1}{M} \int_A \pi(y)\, dy}{\frac{1}{M}} = \int_A \pi(y)\, dy.$$

□

The acceptance probability is $\mathbb{P}(\text{accept}) = 1/M$, so the waiting time follows $\mathrm{Geo}(1/M)$ with mean M. Given N i.i.d. proposals $Y_1, \ldots, Y_N \sim g$ and $U_1, \ldots, U_N \sim \mathrm{Unif}(0, 1)$, with $r_i = \pi(Y_i)/(Mg(Y_i))$, the *RS estimator* of $\mathbb{E}_\pi[h]$ is

$$I^{\mathrm{RS}}_{\pi,N}(h) = \frac{\displaystyle\sum_{i=1}^{N} \mathbf{1}\{U_i \leq r_i\}\, h(Y_i)}{\displaystyle\sum_{i=1}^{N} \mathbf{1}\{U_i \leq r_i\}}. \tag{2.1}$$

Remark 2.3. Rejected proposals are completely discarded. There is no mechanism for moving bad samples toward π; one simply waits for a good draw. When M is large—as it inevitably is in high dimensions—this is very wasteful.

2.3 Importance Sampling

Known target. The key identity is

$$\mathbb{E}_\pi[h] = \int h(x)\, \pi(x)\, dx = \mathbb{E}_g\left[h(Y)\, \frac{\pi(Y)}{g(Y)}\right].$$

Setting $w_i = \pi(Y_i)/g(Y_i)$ and using $\mathbb{E}_g[w_i] = 1$, the *self-normalized IS estimator* is

$$I^{\mathrm{IS}}_{\pi,N}(h) = \frac{\displaystyle\sum_{i=1}^{N} w_i\, h(Y_i)}{\displaystyle\sum_{i=1}^{N} w_i}. \tag{2.2}$$

Unknown normalization (AIS). In practice $\pi = \tilde{\pi}/Z$ with $\tilde{\pi} = e^{-V}$ computable but Z unknown. The unnormalized weights $\tilde{w}_i = \tilde{\pi}(Y_i)/g(Y_i)$ give the *auto-normalized IS (AIS) estimator*

$$I^{\mathrm{AIS}}_{\pi,N}(h) = \frac{\sum_{i=1}^{N} \tilde{w}_i\, h(Y_i)}{\sum_{i=1}^{N} \tilde{w}_i}, \tag{2.3}$$

where Z cancels in the ratio.

Remark 2.4. Unlike RS, IS assigns every proposal a weight rather than accepting or rejecting it. A "bad" proposal (small π/g) contributes little but is not wasted. The cost is high weight variance when π and g differ significantly.

2.4 Asymptotic Theory

Both estimators are ratios of sample averages of i.i.d. variables. Their asymptotics follow from the multivariate CLT and the delta method for $(a, b) \mapsto a/b$.

Theorem 2.5 (CLT for RS and AIS)**.** *Let $h \in L^2(\pi)$. As $N \to \infty$,*

$$\sqrt{N}\big(I^{\mathrm{RS}}_{\pi,N}(h) - \mathbb{E}_\pi[h]\big) \xrightarrow{\mathcal{D}} \mathcal{N}(0,\, G^2_{\mathrm{RS}}), \tag{2.4}$$

$$\sqrt{N}\big(I^{\mathrm{AIS}}_{\pi,N}(h) - \mathbb{E}_\pi[h]\big) \xrightarrow{\mathcal{D}} \mathcal{N}(0,\, G^2_{\mathrm{IS}}), \tag{2.5}$$

where

$$G^2_{\mathrm{RS}} = M \int_{\mathbb{R}^d} \pi(x) \left|h(x) - \mathbb{E}_\pi[h]\right|^2 dx, \tag{2.6}$$

$$G^2_{\mathrm{IS}} = \int_{\mathbb{R}^d} \frac{\pi(x)^2}{g(x)} \left|h(x) - \mathbb{E}_\pi[h]\right|^2 dx. \tag{2.7}$$

Remark 2.6. The factor M in G^2_{RS} is the price of rejection: only a fraction $1/M$ of proposals contribute, inflating the variance by M.

Proposition 2.7 (Importance beats rejection)**.** $G^2_{\mathrm{IS}} \le G^2_{\mathrm{RS}}$.

See Exercise 1(c) for the proof via Cauchy–Schwarz.

2.5 Non-Asymptotic Bound and the Role of χ^2

The CLT describes large-N fluctuations but does not say how large N must be. The following finite-N bound makes the dependence on N explicit and non-asymptotic.

Theorem 2.8 (Non-asymptotic MSE bound)**.** *For $\|h\|_\infty \le 1$,*

$$\mathbb{E}\Big[\big(I^{\mathrm{AIS}}_{\pi,N}(h) - \mathbb{E}_\pi[h]\big)^2\Big] \le \frac{4}{N}\big(\chi^2(\pi \,\|\, g) + 1\big), \tag{2.8}$$

where $\chi^2(\pi \,\|\, g) = \int \pi(x)^2/g(x)\, dx - 1$.

To achieve MSE $\leq \varepsilon^2$, one needs $N \gtrsim (\chi^2(\pi \,\|\, g)+1)/\varepsilon^2$. Since $G_{\mathrm{IS}}^2 \leq 4(\chi^2(\pi \,\|\, g)+1)$ for $\|h\|_\infty \leq 1$, the χ^2-divergence controls both the asymptotic and non-asymptotic efficiency of AIS.

See the next section for more definitions and discussions of χ^2 and KL.

2.6 The Curse of Dimensionality

Example 2.9 (Product measures)**.** *Let $\pi = \pi_1^{\otimes d}$, $g = g_1^{\otimes d}$, $M_1 = \sup_{x\in\mathbb{R}} \pi_1(x)/g_1(x)$, $c_1 = \chi^2(\pi_1 \,\|\, g_1)$. Then*

$$\sup_{x\in\mathbb{R}^d} \frac{\pi(x)}{g(x)} = M_1^d, \qquad \chi^2(\pi \,\|\, g) = (1+c_1)^d - 1.$$

Both grow exponentially *in d: RS needs M_1^d proposals per accepted sample, and AIS needs $N \sim (1+c_1)^d/\varepsilon^2$ samples to achieve MSE $\leq \varepsilon^2$.*

Example 2.10 (Shifted Gaussians)**.** *Let $\pi = \mathcal{N}(m_1, I_d)$, $g = \mathcal{N}(m_2, I_d)$. Completing the square gives*

$$\chi^2(\pi \,\|\, g) = e^{\|m_1-m_2\|^2} - 1, \qquad D_{\mathrm{KL}}(\pi \,\|\, g) = \tfrac{1}{2}\|m_1 - m_2\|^2.$$

For $m_1 - m_2 = (1,\dots,1)^\top$ so $\|m_1 - m_2\|^2 = d$: AIS requires $N \sim e^d/\varepsilon^2$ samples, even though $D_{\mathrm{KL}}(\pi \,\|\, g) = d/2$ grows only linearly.

The χ^2-divergence, not KL, is the correct measure of IS difficulty: a proposal "close" to π in KL may have exponentially large χ^2, rendering IS completely impractical.

The fundamental obstacle. Both RS and IS use a *static* proposal: independent draws from g are accepted or reweighted. There is no mechanism to move bad proposals toward π. This motivates *dynamical* approaches, where a distribution ρ_t evolves from an easy ρ_0 toward π via an ODE or SDE:

$$dX_t = b_t(X_t)\,dt \qquad \text{(deterministic flow)}, \tag{2.9}$$

$$dX_t = b_t(X_t)\,dt + \sigma_t\,dW_t \qquad \text{(stochastic flow)}. \tag{2.10}$$

The rest of the course constructs and analyzes such dynamics for high-dimensional sampling and generative modeling.

3 Probability Divergences

3.1 f-Divergences

Having seen that $\chi^2(\pi \,\|\, g)$ governs IS efficiency, we place it in the broader framework of f-divergences that can be used to measure dissimilarity between two measures.

Definition 3.1 (f-divergence)**.** *Let $f : (0, \infty) \to \mathbb{R}$ be convex with $f(1) = 0$. The* f-divergence *between $P \ll Q$ with densities p, q on $\mathbb{R}^d$ is*

$$D_f(P \,\|\, Q) = \int_{\mathbb{R}^d} f\!\left(\frac{p(x)}{q(x)}\right) q(x)\, dx = \mathbb{E}_Q\left[f\!\left(\frac{p(Y)}{q(Y)}\right)\right]. \tag{3.1}$$

If $P \not\ll Q$, set $D_f(P \,\|\, Q) = +\infty$.

Proposition 3.2 (Basic properties)**.** *(i)* Non-negativity*: $D_f(P \,\|\, Q) \geq 0$.*

(ii) Definiteness*: if f is strictly convex at 1, then $D_f(P \,\|\, Q) = 0 \iff P = Q$.*
Both follow from Jensen: $D_f(P \,\|\, Q) \geq f(\mathbb{E}_Q[p/q]) = f(1) = 0$.

3.2 Special Cases

$f(t)$	Name	Formula
$t \log t$	KL divergence $D_{\mathrm{KL}}(P \,\|\, Q)$	$\int p \log \frac{p}{q}\, dx$
$-\log t$	Reverse KL $D_{\mathrm{KL}}(Q \,\|\, P)$	$\int q \log \frac{q}{p}\, dx$
$(t-1)^2$	χ^2-divergence $\chi^2(P \,\|\, Q)$	$\int \frac{(p-q)^2}{q}\, dx = \int \frac{p^2}{q}\, dx - 1$
$\frac{1}{2}\lvert t-1\rvert$	Total variation $D_{\mathrm{TV}}(P, Q)$	$\frac{1}{2}\int \lvert p-q\rvert\, dx$

3.3 Comparison and High-Dimensional Behavior

Proposition 3.3 (Ordering)**.** *For any $P \ll Q$,*

$$4\, D_{\mathrm{TV}}(P, Q)^2 \leq D_{\mathrm{KL}}(P \,\|\, Q) \leq \log\!\big(1 + \chi^2(P \,\|\, Q)\big) \leq \chi^2(P \,\|\, Q)\,. \tag{3.2}$$

The first inequality is Pinsker's. The second follows from the log-sum inequality (Exercise 4(d)). The third uses $\log(1+t) \leq t$. In particular, χ^2 is strictly stronger than KL, which is stronger than TV.

For product measures $\pi = \pi_1^{\otimes d}$, $g = g_1^{\otimes d}$:

$D_{\mathrm{KL}}(\pi \,\|\, g) = d \cdot D_{\mathrm{KL}}(\pi_1 \,\|\, g_1)$ (linear), $\qquad \chi^2(\pi \,\|\, g) = (1+\chi^2(\pi_1 \,\|\, g_1))^d - 1$ (exponential).

For shifted Gaussians with $\|m_1 - m_2\|^2 = d$:

$$D_{\mathrm{TV}}(\pi, g) = \Theta(1), \qquad D_{\mathrm{KL}}(\pi \,\|\, g) = \tfrac{d}{2}, \qquad \chi^2(\pi \,\|\, g) = e^d - 1.$$

TV and KL give a benign picture; χ^2 reveals the true exponential difficulty for IS.

3.4 Preview: Density Evolution under ODE/SDE Dynamics

Returning to the dynamical approach (2.9)–(2.10), a key question is: how does the density ρ_t of X_t evolve?

Theorem 3.4 (Continuity and Fokker–Planck equations)**.** *Let ρ_t denote the density of X_t.*

(i) ***ODE*** *$dX_t = b_t(X_t)\,dt$: ρ_t satisfies the* continuity equation

$$\partial_t \rho_t + \nabla \cdot (\rho_t\, b_t) = 0. \tag{3.3}$$

(ii) ***SDE*** *$dX_t = b_t(X_t)\,dt + \sigma_t\, dW_t$: ρ_t satisfies the* Fokker–Planck equation

$$\partial_t \rho_t + \nabla \cdot (\rho_t\, b_t) = \tfrac{1}{2}\sigma_t^2\, \Delta \rho_t. \tag{3.4}$$

Proof sketch of (i). For $\phi \in C_c^\infty(\mathbb{R}^d)$,

$$\begin{aligned}\int \phi\, \partial_t \rho_t\, dx &= \tfrac{d}{dt}\mathbb{E}[\phi(X_t)] = \mathbb{E}[\nabla\phi(X_t)\cdot b_t(X_t)] \\ &= \int \nabla\phi \cdot b_t\, \rho_t\, dx = -\int \phi\, \nabla\cdot(\rho_t b_t)\, dx.\end{aligned}$$

Since ϕ is arbitrary, $\partial_t \rho_t = -\nabla \cdot (\rho_t b_t)$. □

Remark 3.5. Equation (3.3) expresses conservation of mass: ρ_t is transported by b_t with no creation or destruction. Equation (3.4) adds diffusion $\frac{\sigma_t^2}{2}\Delta\rho_t$ from the Brownian noise.

The rest of the course constructs b_t, σ_t so that $\rho_t \to \pi$ efficiently in high dimensions. We will discuss how these lead to dynamical algorithms and how these algorithms achieve accuracy in some f-divergence with certain complexity depending on dimensions.

Week 1 Exercises

1. (CLT for RS and AIS.) Prove Theorem 2.5 by the following steps.

(a) Write $I^{\mathrm{RS}}_{\pi,N}(h) = A_N/B_N$ where

$$A_N = \frac{1}{N}\sum_{i=1}^{N} \mathbf{1}\{U_i \le r_i\}\, h(Y_i), \qquad B_N = \frac{1}{N}\sum_{i=1}^{N} \mathbf{1}\{U_i \le r_i\}.$$

Verify the summands are i.i.d. under g. Compute $\mathbb{E}[A_N]$, $\mathbb{E}[B_N]$, and the 2×2 covariance matrix Σ of (A_N, B_N). Apply the multivariate CLT and the delta method with $\varphi(a, b) = a/b$ to derive G^2_{RS} in (2.6).

(b) Repeat for $I^{\mathrm{AIS}}_{\pi,N}(h)$: identify the i.i.d. summands, compute their joint covariance, and apply the delta method to derive G^2_{IS} in (2.7).

(c) Prove $G^2_{\mathrm{IS}} \le G^2_{\mathrm{RS}}$ using Cauchy–Schwarz. Characterize when equality holds.

2. (Non-asymptotic bound for AIS.) Prove Theorem 2.8: for $\|h\|_\infty \le 1$,

$$\mathbb{E}\Big[\big(I^{\mathrm{AIS}}_{\pi,N}(h) - \mathbb{E}_\pi[h]\big)^2\Big] \le \frac{4}{N}\big(\chi^2(\pi \,\|\, g) + 1\big).$$

Hint. Write $I^{\mathrm{AIS}}_{\pi,N}(h) = \tilde{A}_N/\tilde{B}_N$ and note $I^{\mathrm{AIS}}_{\pi,N}(h) - \mathbb{E}_\pi[h] = (\tilde{A}_N - \mathbb{E}_\pi[h] \cdot \tilde{B}_N)/\tilde{B}_N$. Argue $\tilde{B}_N > 0$ a.s., bound the numerator's second moment using $\|h\|_\infty \le 1$, and show $\mathbb{E}_g[\tilde{w}_i^2]/Z^2 = \chi^2(\pi \,\|\, g) + 1$.

3. (χ^2-divergence in high dimensions.)

(a) Let $\pi = \pi_1^{\otimes d}$ and $g = g_1^{\otimes d}$. Show $\chi^2(\pi \,\|\, g) = (1 + \chi^2(\pi_1 \,\|\, g_1))^d - 1$. *Hint.* Use $\int p_1^{\otimes d}(x)^2/g_1^{\otimes d}(x)\, dx = (\int p_1^2/g_1\, dx_1)^d$. Conclude χ^2 is exponential in d while $D_{\mathrm{KL}}(\pi \,\|\, g) = d \cdot D_{\mathrm{KL}}(\pi_1 \,\|\, g_1)$ is linear.

(b) Let $\pi = \mathcal{N}(m_1, I_d)$ and $g = \mathcal{N}(m_2, I_d)$. By completing the square, show $\chi^2(\pi \,\|\, g) = e^{\|m_1 - m_2\|^2} - 1$. For $m_1 - m_2 = (1, \ldots, 1)^\top$, how many AIS samples are needed to achieve MSE $\le \varepsilon^2$?

(c) For the same Gaussian pair, compute $D_{\mathrm{KL}}(\pi \,\|\, g)$ and $D_{\mathrm{TV}}(\pi, g)$ as functions of $\delta = \|m_1 - m_2\|$. For $\delta^2 = d$, compare all three divergences as $d \to \infty$ and explain why $D_{\mathrm{KL}}(\pi \,\|\, g) = O(d)$ is misleadingly optimistic about IS difficulty.

4. (f-divergence properties.)

(a) Prove $D_f(P \,\|\, Q) \ge 0$ using Jensen's inequality. State precisely where convexity and $f(1) = 0$ are used.

(b) Prove that if f is strictly convex at 1, then $D_f(P \,\|\, Q) = 0 \implies P = Q$ as measures.

(c) Verify each row of the table in Section 3 by substituting the given f into Theorem 3.1.

(d) Prove $D_{\mathrm{KL}}(P \,\|\, Q) \le \log(1 + \chi^2(P \,\|\, Q))$. *Hint.* Write $p/q = 1 + (p-q)/q$ and use $\log t \le t - 1$. Deduce $D_{\mathrm{KL}}(P \,\|\, Q) \le \chi^2(P \,\|\, Q)$.

5. (Optimal IS proposal.) Fix target π and h with $\mathbb{E}_\pi[h] = \mu_h$.

(a) Show G^2_{IS} is minimized over densities g with $\mathrm{supp}(\pi) \subseteq \mathrm{supp}(g)$ by $g^*(x) \propto \pi(x)|h(x) - \mu_h|$. *Hint.* Apply Cauchy–Schwarz: $\int \frac{(\pi|h-\mu_h|)^2}{g}\, dx \cdot \int g\, dx \ge (\int \pi|h - \mu_h|\, dx)^2$.

(b) Compute $G^2_{\mathrm{IS}}(g^*)$ in closed form.

(c) Explain why g^* is not directly usable in practice.

(d) When $h \ge 0$, find a proposal $g^\dagger$ achieving $G^2_{\mathrm{IS}}(g^\dagger) = 0$. Why is $g^\dagger$ also impractical?

6. (Continuity and Fokker–Planck equations.)

(a) Give a careful proof of Theorem 3.4(i) using a test function $\phi \in C_c^\infty(\mathbb{R}^d)$ and integration by parts. State which property each step is used.

(b) Show that a stationary density ρ_∞ of (3.3) satisfies $\nabla \cdot (\rho_\infty b) = 0$. For which drift b is $\pi \propto e^{-V}$ stationary?

(c) Prove (3.4) using Itô's formula: $d\phi(X_t) = \nabla\phi \cdot b_t\, dt + \frac{\sigma_t^2}{2}\Delta\phi\, dt + \sigma_t \nabla\phi \cdot dW_t$.

(d) Verify that for the overdamped Langevin SDE $dX_t = -\nabla V(X_t)\, dt + \sqrt{2}\, dW_t$, the stationary distribution of (3.4) is $\pi \propto e^{-V}$.

4 Langevin Dynamics

4.1 The Overdamped Langevin SDE

Recall the fundamental question: given query access to ∇V, how do we construct a drift b_t and noise level σ_t in (2.10) so that $\rho_t \to \pi \propto e^{-V}$?

Intuition from optimization. If we simply want to find the mode $x^* = \arg\min V$, we run gradient descent $\dot{x}_t = -\nabla V(x_t)$, which drives $x_t \to x^*$ exponentially fast under strong convexity. But sampling from π requires exploring the entire distribution, not just its peak. The *overdamped Langevin SDE* adds Brownian noise to gradient descent:

$$dX_t = -\nabla V(X_t)\,dt + \sqrt{2}\,dW_t. \tag{4.1}$$

The drift $-\nabla V$ pushes X_t toward regions of high probability, while the noise $\sqrt{2}\,dW_t$ prevents collapse and drives exploration. The factor $\sqrt{2}$ is chosen so that $\pi \propto e^{-V}$ is exactly stationary, as we now verify.

4.2 Fokker–Planck Equation and Stationarity

Derivation of the Fokker–Planck equation. Let ρ_t denote the density of X_t under (4.1). For any test function $\phi \in C_c^\infty(\mathbb{R}^d)$, we compute $\frac{d}{dt}\mathbb{E}[\phi(X_t)]$ in two ways.

On one hand, differentiating under the integral:

$$\frac{d}{dt}\mathbb{E}[\phi(X_t)] = \int \phi(x)\,\partial_t \rho_t(x)\,dx.$$

On the other hand, Itô's formula applied to (4.1) gives

$$d\phi(X_t) = \underbrace{\nabla\phi(X_t)\cdot(-\nabla V(X_t))\,dt + \Delta\phi(X_t)\,dt}_{\text{bounded variation}} + \underbrace{\sqrt{2}\,\nabla\phi(X_t)\cdot dW_t}_{\text{martingale}}.$$

Taking expectations, the martingale term vanishes, so

$$\frac{d}{dt}\mathbb{E}[\phi(X_t)] = \int \big(-\nabla\phi\cdot\nabla V + \Delta\phi\big)\rho_t\,dx.$$

Integrating by parts (boundary terms vanish since $\phi \in C_c^\infty$):

$$\int (-\nabla\phi\cdot\nabla V)\,\rho_t\,dx = \int \phi\,\nabla\cdot(\rho_t\,\nabla V)\,dx, \qquad \int \Delta\phi\,\rho_t\,dx = \int \phi\,\Delta\rho_t\,dx.$$

Since ϕ is arbitrary, equating the two expressions gives the *Fokker–Planck equation* for Langevin dynamics:

$$\boxed{\partial_t \rho_t = \nabla\cdot(\rho_t\,\nabla V) + \Delta\rho_t.} \tag{4.2}$$

Compact form via the score. Using the *score trick* $\Delta\rho = \nabla\cdot(\nabla\rho) = \nabla\cdot(\rho\,\nabla\log\rho)$, equation (4.2) becomes

$$\partial_t\rho_t = \nabla\cdot\big(\rho_t\,(\nabla V + \nabla\log\rho_t)\big) = \nabla\cdot\Big(\rho_t\,\nabla\log\frac{\rho_t}{\pi}\Big)\,, \tag{4.3}$$

where the last step uses $\nabla V + \nabla\log\rho_t = -\nabla\log\pi + \nabla\log\rho_t = \nabla\log(\rho_t/\pi)$. This form immediately shows that $\rho_t = \pi$ is a stationary solution (the right-hand side vanishes), and that the dynamics transports mass along $-\nabla\log(\rho_t/\pi)$, pushing ρ_t toward π.

Proposition 4.1 (Stationarity of π)**.** *$\pi \propto e^{-V}$ is a stationary solution of (4.2). Equivalently, if $X_0 \sim \pi$ then $X_t \sim \pi$ for all $t \geq 0$.*

Remark 4.2. The compact form (4.3) reveals that Langevin dynamics is a *gradient flow* of $D_{\mathrm{KL}}(\rho_t \,\|\, \pi)$ in the space of probability measures equipped with the Wasserstein-2 metric (Jordan–Kinderlehrer–Otto, 1998). We will revisit this perspective when studying variational inference.

4.3 Mixing Time and the Wasserstein Metric

We now ask: starting from $\rho_0 \neq \pi$, how quickly does $\rho_t \to \pi$? To measure convergence we use the *Wasserstein-2 metric*, which is well-suited to the coupling arguments below and connects naturally to optimal transport.

Definition 4.3 (Wasserstein-2 metric)**.** *For probability measures μ, ν on $\mathbb{R}^d$,*

$$W_2(\mu,\nu) = \Bigg(\inf_{\gamma\in\Gamma(\mu,\nu)} \int \|x-y\|_2^2\, d\gamma(x,y)\Bigg)^{1/2} = \inf_{X\sim\mu,\,Y\sim\nu} \big(\mathbb{E}\|X-Y\|_2^2\big)^{1/2}, \tag{4.4}$$

where $\Gamma(\mu,\nu)$ denotes all couplings of μ and ν. The minimizing coupling is the optimal transport plan.

Example 4.4. *$W_2(\delta_x,\delta_y) = \|x-y\|_2$. For shifted Gaussians, $W_2(\mathcal{N}(m_1,I_d),\mathcal{N}(m_2,I_d)) = \|m_1-m_2\|_2$, growing like $\sqrt{d}$ for typical shifts, whereas $D_{\mathrm{KL}}(\mathcal{N}(m_1,I_d)\,\|\,\mathcal{N}(m_2,I_d)) = \frac{1}{2}\|m_1-m_2\|^2$ grows like d.*

The *mixing time* to tolerance ε in metric D is

$$t_{\mathrm{mix}}^{D}(\varepsilon;\rho_0) = \inf\big\{t\geq 0 : D(\rho_t,\pi)\leq\varepsilon\big\}. \tag{4.5}$$

4.4 Convergence under Strong Convexity

Intuition from optimization. When V is α-strongly convex, the gradient flow ODE $\dot{x}_t = -\nabla V(x_t)$ satisfies

$$\frac{d}{dt}\|x_t - x^*\|_2^2 = 2(x_t - x^*)\cdot(-\nabla V(x_t)) \leq -2\alpha\|x_t - x^*\|_2^2,$$

using $(x-y)\cdot(\nabla V(x) - \nabla V(y)) \geq \alpha\|x-y\|_2^2$ and $\nabla V(x^*) = 0$, giving $\|x_t - x^*\|_2 \leq e^{-\alpha t}\|x_0 - x^*\|_2$. The Langevin SDE extends this contraction to the level of probability

distributions via a *synchronous coupling*: run two copies of (4.1) with the same Brownian motion so that their noise cancels, reducing the problem to the deterministic ODE argument above.

Definition 4.5 (Strong convexity)**.** *$V \in C^2(\mathbb{R}^d)$ is α-strongly convex if $\nabla^2 V(x) \succeq \alpha I_d$ for all $x \in \mathbb{R}^d$.*

Theorem 4.6 (W_2 contraction for Langevin)**.** *Assume V is α-strongly convex. Let ρ_t be the law of X_t solving (4.1) with $X_0 \sim \rho_0$. Then*

$$W_2(\rho_t, \pi) \le e^{-\alpha t}\, W_2(\rho_0, \pi). \tag{4.6}$$

Proof. Run two copies of (4.1) with the same W_t: $X_0 \sim \rho_0$ and $Y_0 \sim \pi$, so $Y_t \sim \pi$ for all t by stationarity. Since both processes share the same noise, the difference $Z_t = X_t - Y_t$ satisfies the *deterministic* ODE

$$\dot{Z}_t = -\nabla V(X_t) + \nabla V(Y_t).$$

Then $\frac{d}{dt}\|Z_t\|_2^2 = 2(X_t - Y_t) \cdot (-\nabla V(X_t) + \nabla V(Y_t)) \le -2\alpha \|Z_t\|_2^2$ by strong convexity, so Gronwall gives $\mathbb{E}\|X_t - Y_t\|_2^2 \le e^{-2\alpha t}\mathbb{E}\|X_0 - Y_0\|_2^2$. Optimizing over the initial coupling (X_0, Y_0) with $X_0 \sim \rho_0$, $Y_0 \sim \pi$ yields (4.6). □

4.5 Initialization and Dimension Dependence

From (4.6), $W_2(\rho_t, \pi) \le \varepsilon$ after $t \ge \frac{1}{\alpha} \log \frac{W_2(\rho_0, \pi)}{\varepsilon}$. It remains to bound $W_2(\rho_0, \pi)$ for a concrete initialization. For α-strongly convex V, a natural choice is $\rho_0 = \delta_{x^*}$.

Lemma 4.7 (Stein's identity)**.** *For $X \sim \pi \propto e^{-V}$ and smooth $f : \mathbb{R}^d \to \mathbb{R}^d$,*

$$\mathbb{E}_\pi[-\nabla V(X) \cdot f(X)] = -\mathbb{E}_\pi[\nabla \cdot f(X)]. \tag{4.7}$$

Proof. Integration by parts: $\int (-\nabla V) \cdot f\, \pi\, dx = \int \nabla \log \pi \cdot f\, \pi\, dx = \int \nabla \pi \cdot f\, dx = -\int \pi\, \nabla \cdot f\, dx$. □

Proposition 4.8 (Initial distance from the mode)**.** *If V is α-strongly convex, then*

$$W_2^2(\delta_{x^*}, \pi) \le \mathbb{E}_\pi[\|X - x^*\|_2^2] \le \frac{d}{\alpha}.$$

Proof. The first inequality uses (x^*, X) with $X \sim \pi$ as a (suboptimal) coupling. For the second, apply (4.7) with $f(x) = x - x^*$ (so $\nabla \cdot f = d$): $\mathbb{E}_\pi[-\nabla V(X) \cdot (X - x^*)] = -d$. Since $\nabla V(x^*) = 0$, strong convexity gives $(\nabla V(X) - \nabla V(x^*)) \cdot (X - x^*) \ge \alpha \|X - x^*\|_2^2$, so $\alpha\, \mathbb{E}_\pi \|X - x^*\|_2^2 \le d$. □

Corollary 4.9 (Mixing time from mode initialization)**.** *Under α-strong convexity, starting from $\rho_0 = \delta_{x^*}$,*

$$t_{\text{mix}}^{W_2}(\varepsilon; \delta_{x^*}) \le \frac{1}{\alpha} \log \frac{\sqrt{d/\alpha}}{\varepsilon}.$$

Remark 4.10 (Continuous time is not an algorithm). The mixing time in Theorem 4.9 is for the *continuous-time* SDE (4.1), which requires simulating the stochastic process exactly and is not directly implementable. In practice one must discretize (4.1) in time with step size $h > 0$, yielding the *Unadjusted Langevin Algorithm* (ULA):

$$X_{k+1} = X_k - h\,\nabla V(X_k) + \sqrt{2h}\,\xi_k, \qquad \xi_k \overset{\text{i.i.d.}}{\sim} \mathcal{N}(0, I_d).$$

The true measure of algorithmic complexity is therefore the number of gradient evaluations k needed to reach $W_2(\mathrm{law}(X_k), \pi) \leq \varepsilon$. This discrete-time analysis, which gives a practically meaningful complexity bound, is the subject of the next lecture. But just from the structure of the bound, the complexity will not scale with dimension exponentially, when the target distribution is strongly log-concave, i.e., V is strongly convex.

5 Langevin Monte Carlo

6 Score-Based Diffusion Models

7 Flow Matching, Rectified Flow, and Stochastic Interpolants